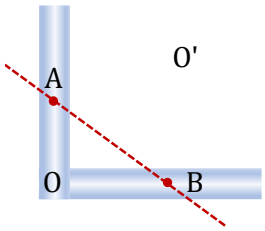
RACE
Regular Analysis Through Continuous Exercise

Center of Mass, Momentum and Collision : RACE-01

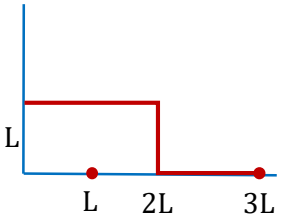
1. Figure shows two cylindrical rods whose center of mass is marked as *A* and *B*. Line *AB* divides the region in two parts one containing point *O* (region 1) and other containing point *O*' (region 2). Choose the correct option regarding the center of mass of the combined system.



(A) The center of mass of the system lies in region 1.
(B) The center of mass of the system lies in region 2.
(C) The center of mass of the system lies on line *AB*.
(D) The center of mass of the system may lie in region 1 or region 2 depending on the mass of the rods.

PCM211

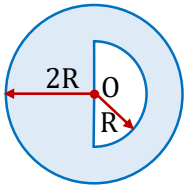
2. The position vector of the centre of mass $\vec{r}cm$ of an symmetric uniform bar of negligible area of cross-section as shown in figure is :



(A) $\vec{r}cm = \frac{13}{8}L\hat{x} + \frac{5}{8}L\hat{y}$ (B) $\vec{r}cm = \frac{11}{8}L\hat{x} + \frac{3}{8}L\hat{y}$
(C) $\vec{r}cm = \frac{3}{8}L\hat{x} + \frac{11}{8}L\hat{y}$ (D) $\vec{r}cm = \frac{5}{8}L\hat{x} + \frac{13}{8}L\hat{y}$

PCM212

3. From a uniform disc of radius $2R$, a semi circular disc of radius R is cut and removed. Find out the distance of centre of mass of this part from the point '*O*' as shown in the figure is:



(A) $\frac{3R}{28\pi}$ (B) $\frac{3R}{4\pi}$ (C) $\frac{4R}{21\pi}$ (D) none of these

PCM213

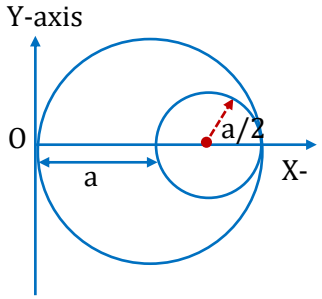
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4. A rod of length L has non-uniform linear mass density given by $\rho(x) = a + b\left(\frac{x}{L}\right)^2$, where a and b are constants and $0 \le x \le L$. The value of x for the centre of mass of the rod is at :

(A) $\frac{4}{3}\left(\frac{a+b}{2a+3b}\right)L$ (B) $\frac{3}{2}\left(\frac{a+b}{2a+b}\right)L$ (C) $\frac{3}{2}\left(\frac{2a+b}{3a+b}\right)L$ (D) $\frac{3}{4}\left(\frac{2a+b}{3a+b}\right)L$

PCM214

5. A circular hole of radius $\left(\frac{a}{2}\right)$ is cut out of a circular disc of radius 'a' as shown in figure. The centroid of the remaining circular portion with respect to point 'O' will be:



(A) $\frac{1}{6}a$ (B) $\frac{10}{11}a$ (C) $\frac{5}{6}a$ (D) $\frac{2}{3}a$

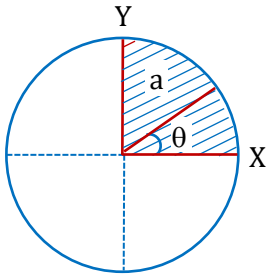
PCM215

6. The disc of mass M with uniform surface mass density σ is shown in the figure. The centre of mass of the quarter disc (the shaded area) is at the position $\frac{x}{3}\frac{a}{\pi}, \frac{x}{3}\frac{a}{\pi}$ where x is _____.

(Round off to the Nearest Integer)

[a is an area as shown in the figure]

Rankers Academy JEE



PCM216

7. Two bodies of mass 1 kg and 3 kg have position vectors $\hat{\imath} + 2\hat{\jmath} + \hat{k}$ and $-3\hat{\imath} - 2\hat{\jmath} + \hat{k}$ respectively. The magnitude of position vector of centre of mass of this system will be similar to the magnitude of vector :

(A) $\sqrt{6}$ (B) $\sqrt{5}$ (C) $\sqrt{7}$ (D) $\sqrt{8}$

PCM217

8. Two blocks of masses 10 kg and 30 kg are placed on the same straight line with coordinates (0, 0) cm and (x, 0) cm respectively. The block of 10 kg is moved on the same line through a distance of 6 cm towards the other block. The distance through which the block of 30 kg must be moved to keep the position of centre of mass of the system unchanged is :

(A) 4 cm towards the 10 kg block (B) 2 cm away from the 10 kg block

(C) 2 cm towards the 10 kg block (D) 4 cm away from the 10 kg block

PCM218

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9. The distance of centre of mass from end A of a one-dimensional rod (AB) having mass density $\lambda = \lambda_0\left(1 - \frac{x^2}{L^2}\right) kg/m$ and length L (in meter) is $\frac{3L}{\alpha}\ m$. The value of α is .....

(where x is the distance form end A)

PCM219

10. Three identical spheres each of mass M are placed at the corners of a right-angled triangle with mutually perpendicular sides equal to $3\ m$ each. Taking point of intersection of mutually perpendicular sides as origin, the magnitude of position vector of centre of mass of the system will be $\sqrt{x}m$. The value of x is

PCM220

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ANSWER KEY

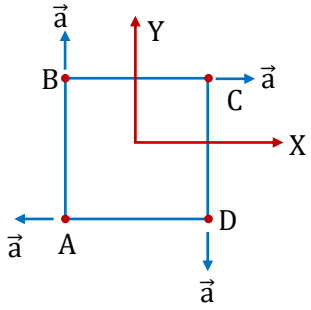
Question	1	2	3	4	5	6	7	8	9	10
Answer	C	A	C	D	C	4	A	C	8	2

For More Join: @IITJEE_Advanced

RACE
Regular Analysis Through Continuous Exercise

Center of Mass, Momentum and Collision : RACE-02

1. Four particles A, B, C and D with masses $m_A = m$, $m_B = 2m$, $m_C = 3m$ and $m_D = 4m$ are at the corners of a square. They have accelerations of equal magnitude with directions as shown. The acceleration of the centre of mass of the particles is :



(A) $\frac{a}{5}(\hat{i}-\hat{j})$ (B) $\frac{a}{5}(\hat{i}+\hat{j})$ (C) Zero (D) $a(\hat{i}+\hat{j})$

PCM221

2. A particle of mass m is projected with a speed u from the ground at an angle $\theta = \frac{\pi}{3}$ w.r.t. horizontal (x-axis). When it has reached its maximum height, it collides completely inelastically with another particle of the same mass and velocity $u\,\hat{i}$. The horizontal distance covered by the combined mass before reaching the ground is:

(A) $\frac{3\sqrt{2}}{4}\frac{u^2}{g}$ (B) $2\sqrt{2}\frac{u^2}{g}$ (C) $\frac{3\sqrt{3}}{8}\frac{u^2}{g}$ (D) $\frac{5}{8}\frac{u^2}{g}$

PCM222

3. A bullet of '$4g$' mass is fired from a gun of mass 4 kg. If the bullet moves with the muzzle speed of 50 ms^{-1}, the impulse imparted to the gun and velocity of recoil of gun are :

(A) 0.4 $kg\ ms^{-1}$, 0.1 ms^{-1} (B) 0.2 $kg\ ms^{-1}$, 0.05 ms^{-1}
(C) 0.2 $kg\ ms^{-1}$, 0.1 ms^{-1} (D) 0.4 $kg\ ms^{-1}$, 0.05 ms^{-1}

PCM223

4. A ball of mass 0.15 kg hits the wall with its initial speed of 12 ms^{-1} and bounces back without changing its initial speed. If the force applied by the wall on the ball during the contact is 100 N. calculate the time duration of the contact of ball with the wall.

(A) 0.018 s (B) 0.036 s (C) 0.009 s (D) 0.072 s

PCM224

5. A body of mass 8 kg and another of mass 2 kg are moving with equal kinetic energy. The ratio of their respective momenta will be :

(A) 1 : 1 (B) 2 : 1 (C) 1 : 4 (D) 4 : 1

PCM225

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6. A body of mass M at rest explodes into three pieces, in the ratio of masses 1 : 1 : 2. Two smaller pieces fly off perpendicular to each other with velocities of 30 ms^{-1} and 40 ms^{-1} respectively. The velocity of the third piece will be :

(A) 15 ms^{-1} (B) 25 ms^{-1} (C) 35 ms^{-1} (D) 50 ms^{-1}

PCM226

7. A machine gun of mass 10 kg fires 20 g bullets at the rate of 180 bullets per minute with a speed of 100 ms^{-1} each. The recoil velocity of the gun is :

(A) 0.02 m/s (B) 2.5 m/s (C) 1.5 m/s (D) 0.6 m/s

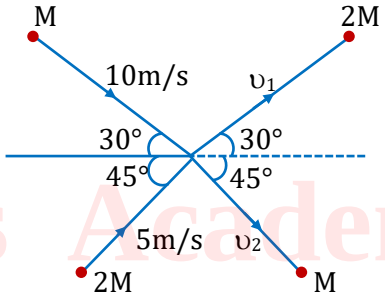
PCM227

8. 100 balls each of mass m moving with speed v simultaneously strike a wall normally and reflected back with same speed, in time t s. The total force exerted by the balls on the wall is

(A) $\frac{100mv}{t}$ (B) $\frac{200mv}{t}$ (C) 200 mvt (D) $\frac{mv}{100t}$

PCM228

9. Two particles, of masses M and $2M$, moving, as shown, with speeds of 10 m/s and 5 m/s, collide elastically at the origin. After the collision, they move along the indicated directions with speeds v_1 and v_2, respectively. The values of v_1 and v_2 are nearly :



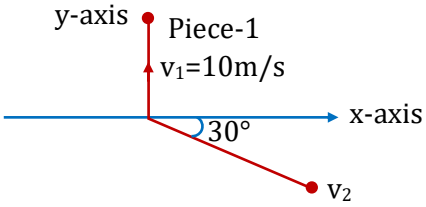
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(A) 3.2 m/s and 6.3 m/s (B) 3.2 m/s and 12.6 m/s

(C) 6.5 m/s and 6.3 m/s (D) 6.5 m/s and 3.2 m/s

PCM229

10. A ball of mass 10 kg moving with a velocity $10\sqrt{3}$ m/s along the x-axis, hits another ball of mass 20 kg which is at rest. After the collision, first ball comes to rest while the second ball disintegrates into two equal pieces. One piece starts moving along y-axis with a speed of 10 m/s. The second piece starts moving at an angle of 30° with respect to the x-axis. The velocity of the ball moving at 30° with x-axis is x m/s. The configuration of pieces after collision is shown in the figure below. The value of x to the nearest integer is _____ .



PCM230

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	A	C	B	B	B	B	D	B	C	20

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RACE
Regular Analysis Through Continuous Exercise

Center of Mass, Momentum and Collision : RACE-03

1. A simple pendulum, made of a string of length l and a bob of mass m, is released from a small angle θ_0. It strikes a block of mass M, kept on a horizontal surface at its lowest point of oscillations, elastically. It bounces back and goes up to an angle θ_1. Then M is given by :

(A) $\frac{m}{2}\left(\frac{\theta_0-\theta_1}{\theta_0+\theta_1}\right)$ (B) $\frac{m}{2}\left(\frac{\theta_0+\theta_1}{\theta_0-\theta_1}\right)$ (C) $m\left(\frac{\theta_0+\theta_1}{\theta_0-\theta_1}\right)$ (D) $m\left(\frac{\theta_0-\theta_1}{\theta_0+\theta_1}\right)$

PCM231

2. A body A, of mass $m = 0.1\ kg$ has an initial velocity of $3\hat{\imath}\ ms^{-1}$. It collides elastically with another body, B of the same mass which has an initial velocity of $5\hat{\jmath}\ ms^{-1}$. After collision, A moves with a velocity $\vec{v} = 4(\hat{\imath}+\hat{\jmath})$. The energy of B after collision is written as $\frac{x}{10}J$. The value of x is _____.

PCM232

3. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body 'P' having mass M moving with speed 'u' has head-on collision elastically with another body 'Q' having mass 'm' initially at rest. If $m << M$, body 'Q' will have a maximum speed equal to '$2u$' after collision.

Reason R : In an elastic collision, the momentum and kinetic energy are both conserved.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is not correct but R is correct.

(B) Both A and R are correct but R is NOT the correct explanation of A.

(C) Both A and R are correct and R is the correct explanation of A.

(D) A is correct but R is not correct.

PCM233

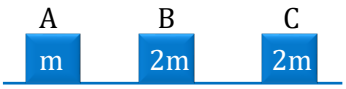
4. A rubber ball is released from a height of 5 m above the floor. It bounces back repeatedly, always rising to $\frac{81}{100}$ of the height through which it falls. Find the average speed of the ball.

(Take $g = 10\ ms^{-2}$)

(A) 3.0 ms^{-1} (B) 3.50 ms^{-1} (C) 2.0 ms^{-1} (D) 2.50 ms^{-1}

PCM234

5. Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. The masses of A, B and C are m, 2 m and 2 m respectively. A moves towards B with a speed of 9 m/s and makes an elastic collision with it. There after B makes a completely inelastic collision with C. All motions occur along same straight line. The final speed of C is :



(A) 6 m/s (B) 9 m/s (C) 4 m/s (D) 3 m/s

PCM235

For More Join: @IITJEE_Advanced

6. A body of mass M moving at speed V_0 collides elastically with a mass 'm' at rest. After the collision, the two masses move at angles θ_1 and θ_2 with respect to the initial direction of motion of the body of mass M. The largest possible value of the ratio M/m, for which the angles θ_1 and θ_2 will be equal, is

(A) 4 (B) 1 (C) 3 (D) 2

PCM236

7. Two billiard balls of mass 0.05 kg each moving in opposite directions with 10 ms^{-1} collide and rebound with the same speed. If the time duration of contact is t = 0.005 s, then what is the force exerted on the ball due to each other?

(A) 100 N (B) 200 N (C) 300 N (D) 400 N

PCM237

8. A man of 60 kg is running on the road and suddenly jumps into a stationary trolly car of mass 120 kg. Then. the trolly car starts moving with velocity 2 ms^{-1}. The velocity of the running man was _____ ms^{-1}. when he jumps into the car.

PCM238

9. A pendulum of length 2 m consists of a wooden bob of mass 50 g. A bullet of mass 75 g is fired towards the stationary bob with a speed v. The bullet emerges out of the bob with a speed $\frac{v}{3}$ and the bob just completes the vertical circle. The value of v is _____ ms^{-1}. (If g = 10 m/s^2)

PCM239

10. What percentage of kinetic energy of a moving particle is transferred to a stationary particle when it strikes the stationary particle of 5 times its mass?
(Assume the collision to be head-on elastic collision)

(A) 50.0% (B) 66.6% (C) 55.5% (D) 33.3%

PCM240

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	C	1	C	D	D	C	B	6	10	C

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Regular Analysis Through Continuous Exercise

Center of Mass, Momentum and Collision : RACE-04

1. Figures (a), (b), (c) and (d) show variation of force with time.

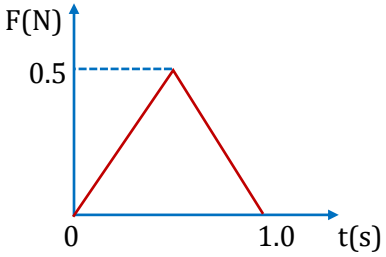


Fig (a)

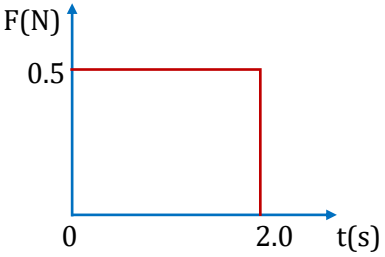


Fig (b)

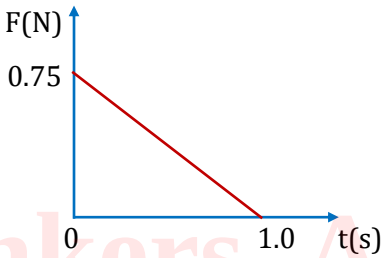


Fig (c)

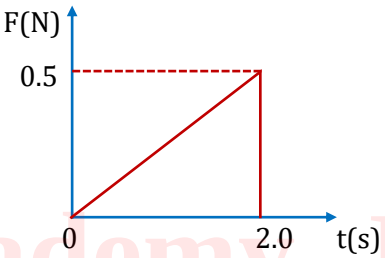


Fig (d)

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The impulse is highest in figure.

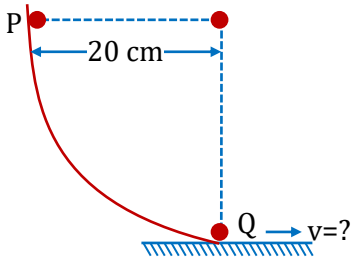
(A) Fig (c) (B) Fig (b) (C) Fig (a) (D) Fig (d)

PCM241

2. A body of mass 1 kg collides head on elastically with a stationary body of mass 3 kg. After collision, the smaller body reverses its direction of motion and moves with a speed of 2 m/s. The initial speed of the smaller body before collision is _____ ms^{-1}.

PCM242

3. As per the given figure, a small ball P slides down the quadrant of a circle and hits the other ball Q of equal mass which is initially at rest. Neglecting the effect of friction and assume the collision to be elastic, the velocity of ball Q after collision will be : (g = 10 m/s^2)



(A) 0 (B) 0.25 m/s (C) 2 m/s (D) 4 m/s

PCM243

For More Join: @IITJEE_Advanced

4. A ball of mass 200 g rests on a vertical post of height 20 m. A bullet of mass 10 g, travelling in horizontal direction, hits the centre of the ball. After collision both travels independently. The ball hits the ground at a distance 30 m and the bullet at a distance of 120 m from the foot of the post. The value of initial velocity of the bullet will be (If g = 10 m/s^2)

(A) 120 m/s (B) 60 m/s (C) 400 m/s (D) 360 m/s

PCM244

5. A ball is dropped from a height of 20 m. If the coefficient of restitution for the collision between ball and floor is 0.5, after hitting the floor, the ball rebounds to a height of_____m.

PCM245

6. **Statement-1 :** In a perfectly inelastic collision there is a limit to the loss of kinetic energy of colliding bodies.

Statement-2 : In perfectly inelastic collision, linear momentum of system is conserved.

(A) Statement-1 is True, Statement-2 is True ; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True ; Statement-2 is not a correct explanation for Statement-1.

(C) Statement-1 is True, Statement-2 is False.

(D) Statement-1 is False, Statement-2 is True.

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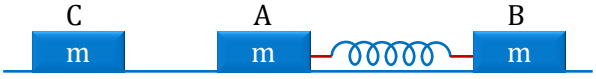
PCM246

7. A spaceship in space sweeps stationary interplanetary dust. As a result, its mass increases at a rate $\frac{dM(t)}{dt} = bv^2(t)$, where $v(t)$ is its instantaneous velocity. The instantaneous acceleration of the satellite is:

(A) $-\frac{2bv^3}{M(t)}$ (B) $-\frac{bv^3}{2M(t)}$ (C) $-bv^3(t)$ (D) $-\frac{bv^3}{M(t)}$

PCM247

8. Two identical blocks A and B each of mass m resting on the smooth horizontal floor are connected by a light spring of natural length L and spring constant K. A third block C of mass m moving with a speed v along the line joining A and B collides with A. The maximum compression in the spring is

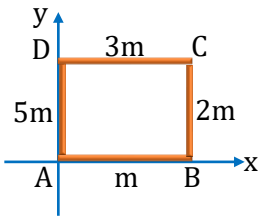


(A) $v\sqrt{\frac{M}{2K}}$ (B) $\sqrt{\frac{mv}{2K}}$ (C) $\sqrt{\frac{mv}{K}}$ (D) $\sqrt{\frac{m}{2K}}$

PCM248

For More Join: @IITJEE_Advanced

9. Four rods of equal lengths AB, BC, CD and DA have mass m, $2m$, $3m$ and $5m$ respectively are placed in x-y plane (see figure) given $AB = BC = CD = DA = 4$ meters. The centre of mass of the system is satisfy (x and y are in meters)



(A) $x > 4, y > 4$ (B) $x < 4, y < 4$ (C) $x = 0, y > 2$ (D) $x > 2, y = 0$

PCM249

10. Two blocks of mass $1kg$ and $2\ kg$ are travelling in the opposite direction with velocity of $3m/s$ and $6\ m/s$ respectively. What is their total kinetic energy (in J) in centre of mass frame of reference ?

PCM250

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ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	4	C	D	5	B	D	A	B	27 J

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